

Introduction to Quantum Mechanics for Quantum Computing

Lecture 1: States, Measurement, and Unitary Evolution (Classical Bits to Qubits)

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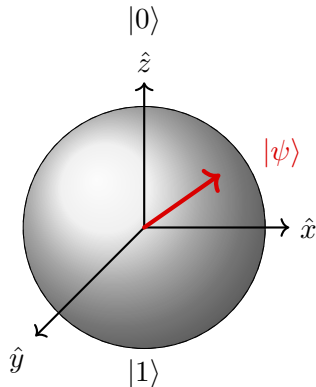
What Is a Hilbert Space?

- A Hilbert space is a **vector space** with:
 - Complex scalars
 - An inner product $\langle \phi | \psi \rangle$
 - Completeness (limits of Cauchy sequences exist)
- The inner product defines:
 - Length: $\langle \psi | \psi \rangle$
 - Angles (orthogonality)
 - Probabilities via Born's rule
- For quantum computing:

$$\mathcal{H}_{1 \text{ qubit}} = \mathbb{C}^2 \quad \mathcal{H}_{n \text{ qubits}} = \mathbb{C}^{2^n}$$



Bloch Sphere (Geometric Picture of Pure Qubit States)



- Any pure qubit state can be written:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

- θ, ϕ parametrize a point on the sphere.
- Gates often look like **rotations** of this vector.



Two States That Look Identical in Z-Measurement

Consider:

$$|\psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle, \quad |\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

- In the Z-basis, both give $p(0) = p(1) = \frac{1}{2}$.
- But they are **different states** (different relative phase).

Concept check: Are these the same state since Z-measurement statistics match?



Postulate 4: Composite Systems

Postulate 4 (Tensor product rule)

If systems have spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the joint system lives in $\mathcal{H}_A \otimes \mathcal{H}_B$.

- Two qubits live in $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$.
- Basis: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- General 2-qubit pure state:

$$|\psi\rangle = \sum_{x \in \{0,1\}^2} \psi_x |x\rangle, \quad \sum_x |\psi_x|^2 = 1.$$

Concept check: How many amplitudes does an n -qubit state have?



Operators on Composite Systems

- If a gate acts only on one subsystem, tensor it with identities.
- Example: apply X to the second qubit of a 3-qubit state:

$$X_2 = I \otimes X \otimes I.$$

- Operators on different subsystems commute:

$$(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I).$$



Preview: Entanglement (Next Time)

- Some joint states cannot be written as $|\psi_A\rangle \otimes |\psi_B\rangle$.
- Example (Bell state):

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

- These non-separable correlations are called **entanglement** and are a key resource in quantum information.



Summary (Lecture 1)

- **States are vectors** in a complex Hilbert space; a qubit is $\mathbb{C}^2$.
- **Measurement** gives probabilities (Born rule) and updates the state (collapse in the chosen basis).
- **Unitary evolution** preserves probability and is reversible; in physics it is generated by H via Schrödinger evolution.
- **Relative phase matters**: it changes interference outcomes after a basis change.
- **Composite systems** use tensor products; this sets up entanglement for next time.

